Predicting Sediment Particle Size Distributions Using Machine Learning

Random Forest and Support Vector Regression, a replication of Egan (Stanford CS229)

UE24CS352A – Machine Learning | Mini-Project Review

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github.com/Siddhesh249/Predicting-Sediment-Particle-Size-Distributions

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Why Predict Sediment Particle Size?

- Suspended marine sediment controls coastal erosion, contaminant transport and carbon sequestration.
- Particle settling velocity is the key parameter in sediment-transport models, but it is poorly known.
- Particle size changes by orders of magnitude within seconds through **flocculation** (aggregation and break-up).
- No general semi-empirical model predicts size accurately.

Our task

Supervised regression: 8 measurable inputs → 2 targets

Target 1: d_{50} (median particle diameter)

Target 2: σ^2 (variance of the size distribution)

Success metric: R^2 on unseen test data

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Replicating Egan (Stanford CS229, 2019)

Why synthetic? The original field dataset is not publicly available, so we built a physically motivated generator that reproduces the variables and nonlinear relationships.

	ORIGINAL PAPER	OUR REPLICATION
Data	Real field data, South San Francisco Bay (LISST-100x and ac-9 instruments)	Synthetic data with the same variables and structure
Samples	1,648 hourly examples	1,648 hourly examples
Random Forest	32 trees	400 trees (OOB score plateaus early)
SVR tuning	C = 2048, ε = 4	Own 5-fold grid search: C = 100, ε = 4
SVR inputs	Top-4 features: S, u_b, n_p, T	Same top-4 features
Result	$R^2 > 0.75$ on test sets	R^2 between 0.84 and 0.92

G. Egan, "Machine Learning for Predicting Sediment Particle Size Distributions", Stanford CS229, 2019.

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Dataset: 8 Inputs, 2 Targets

SYMBOL	MEANING	SIMULATED AS
S	Salinity	Uniform(20, 35)
u_b	Wave-orbital velocity	AR(1) process, mean 0.25
n_p	Particle index of refraction	Normal(1.38, 0.03^2)
T	Water temperature (°C)	$16 + 6 \cdot \sin(2\pi t/360)$ + noise
a676/a650	Algal (organic) absorption peak	Normal(1.0, 0.18^2)
a450/a676	Detrital (inorganic) absorption peak	Normal(0.9, 0.2^2)
chl-a	Chlorophyll-a (organic proxy)	Lognormal, median 0.8
u	Mean tidal / flow velocity	AR(1) process, mean 0.18

Targets

- $\mathbf{d_{50}}$ (median particle diameter)
- $\boldsymbol{\sigma^2}$ (size-distribution variance)

Dataset size: **1,648 hourly samples** from 2018-07-01, fixed random seed 42.

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From Raw Data to Train/Test Split

Nonlinear target generation

$$d_{50} = 5S + 300 \cdot u_b + 90 \cdot n_p + 2T + 30 \cdot z_S \cdot (u_b - \text{mean}\, u_b) + 12 \cdot \ln(1 + \text{chl-a}) + 40 \cdot (a676/a650 - 1)^2 + \text{noise}$$

$$\sigma^2 = 0.12 \cdot d_{50}^{1.25} + 8u + 20 \cdot (a450/a676 - 0.9) + \text{noise}$$

z_S is standardised salinity. Realism added on purpose: noise, **2% missing feature values**, **0.5% target outliers**.

Generate

– Dataset_Generator.py → Dataset.csv (1,648 rows)

Drop missing values

→ 1,402 rows

Remove outliers

(above the 99th percentile of either target) → **1,373 rows**

Split 70/30

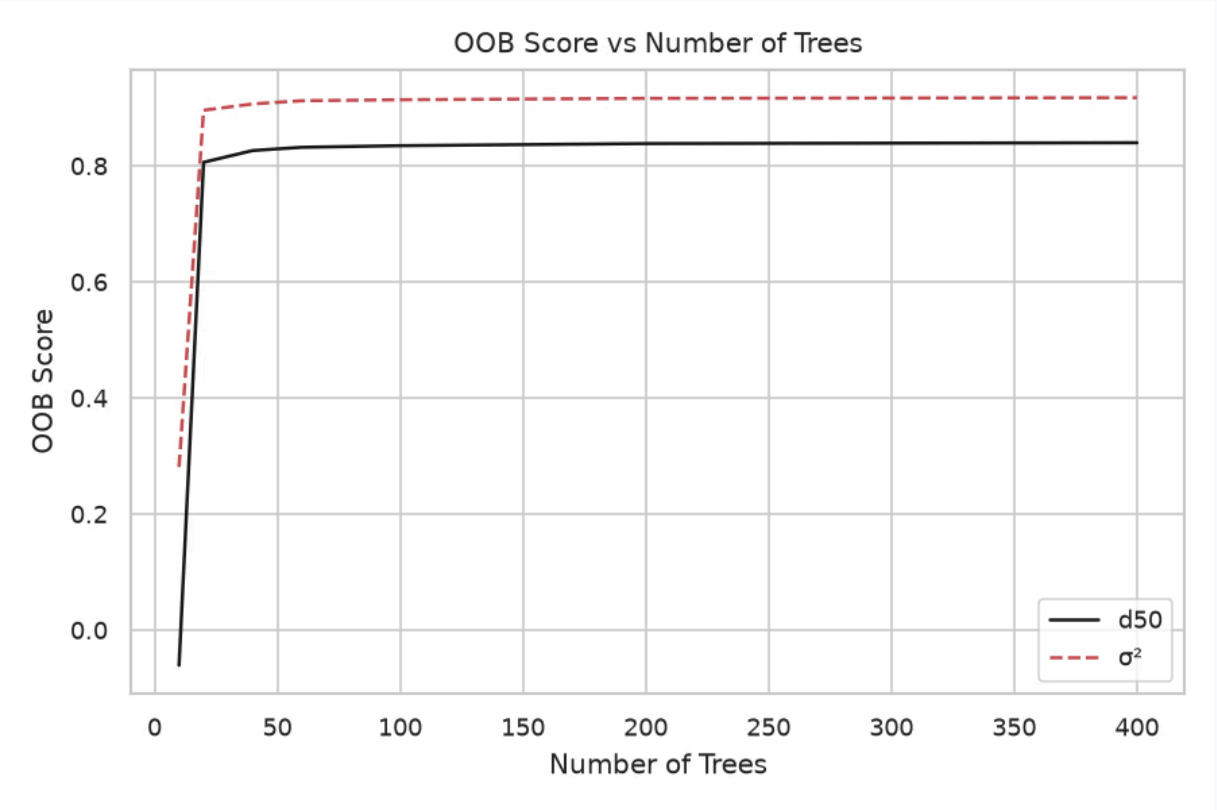
→ **961 train / 412 test** (random_state = 42)

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Random Forest: Averaging Many Trees

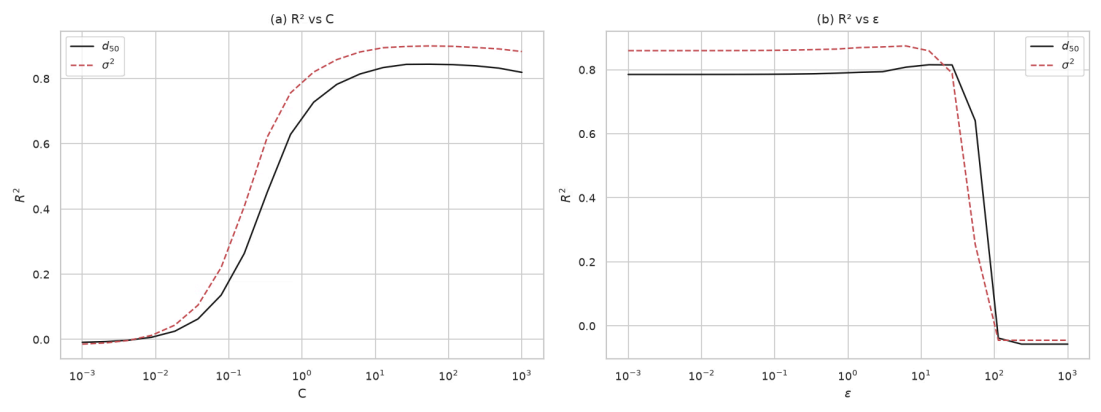
$$\hat{y}(x) = \frac{1}{B} \cdot \sum_{b=1}^{B} T_b(x), \quad b = 1 \dots B$$

- Average of B decision trees, each trained on a bootstrap sample.
- Captures nonlinear interactions and resists overfitting.
- Setup: **B = 400 trees, all 8 features**.
- **Out-of-bag (OOB) score**: an internal, unbiased accuracy estimate used to choose B; it levels off after about 60 trees.



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SVR: Margin-Based Regression with an RBF Kernel



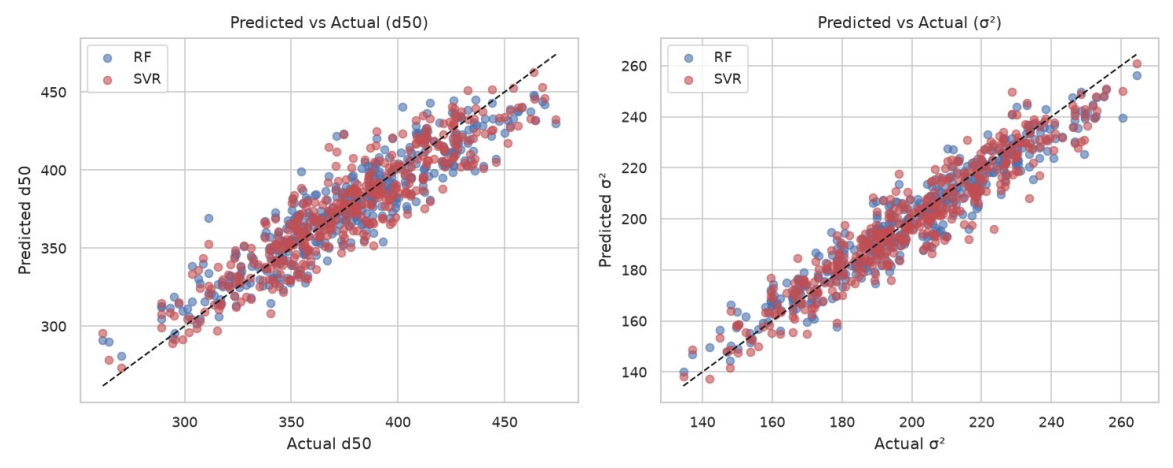
$$\text{minimise } \frac{1}{2}\|w\|^2 + C \cdot \sum_i (\xi_i + \xi_i^*)$$

$$K(x, x') = \exp(-\gamma \|x - x'\|^2)$$

- Errors inside an ε-tube are ignored; C trades flatness against violations.
- Inputs: **only the top-4 features (S, u_b, n_p, T)**, standardised, as in the paper.
- 5-fold grid search over C ∈ {1, 10, 100, 500, 1000, 2048, 3000} and ε ∈ {0.1, 0.5, 1, 2, 4} → best **C = 100, ε = 4**.

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Both Models Exceed the 0.75 Benchmark



$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}$$

MODEL	INPUTS	R^2 FOR D_{50}	R^2 FOR Σ^2
Random Forest (B = 400)	all 8 features	0.840	**0.915**
SVR (RBF, C = 100, ε = 4)	S, u_b, n_p, T	**0.842**	0.903

The two models are effectively tied on d_{50}; Random Forest is slightly better on σ^2.

Test set, 412 samples. Points close to the dashed y = x line indicate accurate predictions.

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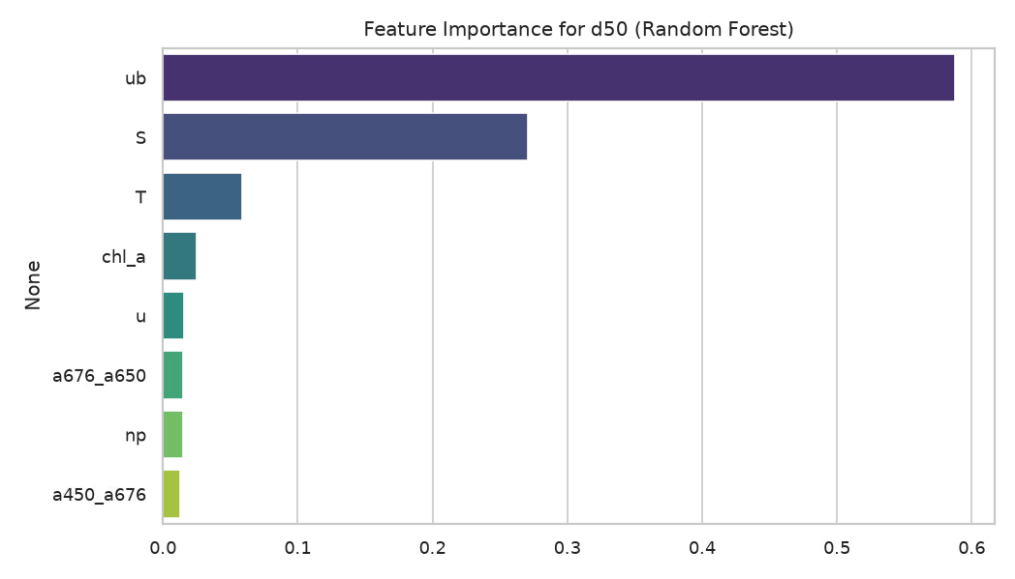
What We Learned

Key insights

Wave-orbital velocity (0.59) and salinity (0.27) carry about 86% of the importance for d_{50}.

SVR matches Random Forest using only 4 of 8 features.

SVR R^2 rises sharply for $C \gtrsim 1$ and collapses when ε exceeds about 30.



Limitations

Data are synthetic, so results validate the pipeline, not real-world accuracy.

A single random split of autocorrelated hourly data may be optimistic.

σ^2 scores higher partly because it is generated from d_{50}.

Conclusion and future work

Data-driven models are a viable alternative to semi-empirical flocculation formulas.

Next: real LISST/ac-9 data, time-blocked cross-validation, gradient boosting and neural-network baselines.

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Thank You

Questions?

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